

L11a: Markov Models and Markov Chains

The Markov property, the transition matrix, and the stationary distribution

CHEME 5800 Cornell University

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Objectives for Today

How do we model a random system whose next state depends only on where it is now?

Today's concepts

- Describe the Markov property: the probability of the next state depends only on the present state, which makes random systems much easier to model and analyze.
- Build a transition matrix: hold the probability of moving from each state to every state in one time step, and state the properties every row must satisfy.
- Compute the stationary distribution: find the distribution that the transition matrix leaves unchanged, using power iteration.

Lecture: Foundations of Markov Models and Markov Chains

Lecture and Example

Lecture: Foundations of Markov Models and Markov Chains
In this lecture, we define the Markov property, collect the one-step transition probabilities in a matrix, and compute the stationary distribution by power iteration.

Example: Properties of Markov Models and Stationary Distributions
In this example, we build a three-state Markov chain, compute its stationary distribution two ways, and check the predictions by simulation.

Markov models capture real-world (stochastic) dynamics well when the present state contains all relevant information.

Discrete Markov Model States

A discrete Markov model is a stochastic (random) process that moves between states with defined probabilities. It is determined by the tuple

$$\mathcal{M} = (\mathcal{S}, \mathbf{P}),$$

where $\mathcal{S}$ is the set of states and $\mathbf{P}$ is the transition matrix.

The state space $\mathcal{S}$ is the set of all values the system can assume. What the states represent depends on the problem:

- Letters in words, $\mathcal{S} = \{a, b, c, \ldots, z\}$: generate words that start with a specific letter, one character at a time.
- Investor moods, $\mathcal{S} = \{\text{bullish}, \text{neutral}, \text{bearish}\}$: simulate how an investor's mood changes as they watch the market and the news.

The Markov Property

Let X_n be the state of a stochastic process at time step n. The process has the Markov property if, for every $n \geq 0$ and every state history with positive probability,

$$P(X_{n+1} = s_{n+1} \mid X_0 = s_0, \ldots, X_n = s_n) = P(X_{n+1} = s_{n+1} \mid X_n = s_n).$$

Given the present state, the next state is conditionally independent of the past states. We can update the state distribution without storing the full history of the process.

One-Step Transition Probabilities

For a finite state space $\mathcal{S} = \{s_1, \dots, s_N\}$, the probability of moving from state s_i to state s_j in one step is

$$p_{ij} = P(X_{n+1} = s_j \mid X_n = s_i).$$

We assume the process is time-homogeneous: these probabilities stay constant over time, so p_{ij} does not depend on n, and the same transition matrix applies at every step.

With this representation, matrix operations describe the chain's long-run behavior.

Why the Markov Property Matters

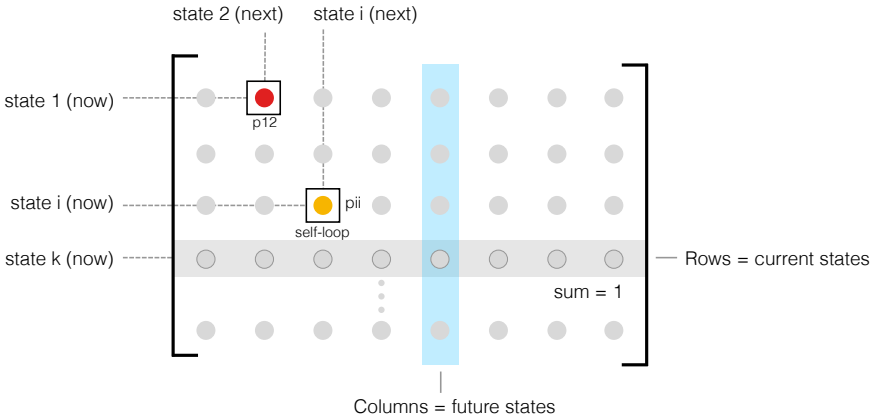
The Markov property simplifies our modeling work: instead of tracking the entire history of a process, we only need to know the present state.

- Without this property, the number of possible historical sequences would grow exponentially.
- With it, the problem becomes tractable and easier to work with.

The Markov property is an assumption about the real world, and not all stochastic processes satisfy it exactly. By choosing the state representation carefully to capture the relevant information, we can often make the Markov assumption work well in practice.

The Transition Matrix

The matrix $\mathbf{P} \in \mathbb{R}^{N \times N}$ collects the one-step probabilities; row i is the next-state distribution from s_i:



Properties of the Transition Matrix

Every transition matrix $\mathbf{P}$ has the following properties:

- Every entry is non-negative: $p_{ij} \geq 0$.
- Rows represent the current states, and columns represent the next states (our convention). Entry p_{ij} is the probability of moving from s_i to s_j.
- Every row sums to one:

$$\sum_{j=1}^{N} p_{ij} = 1.$$

  Starting from any state, the chain must move to one of the states in $\mathcal{S}$, including the possibility of staying where it is.

The Stationary Distribution

Each distribution π is a row vector whose j-th element is the probability that the chain is in state s_j. The stationary distribution $\bar{\pi}$ remains unchanged as the system evolves:

$$\boxed{\bar{\pi} = \bar{\pi}\,\mathbf{P}}$$

This is analogous to the steady state of a physical system. In an ergodic chain (every state can be reached from every other state, and the chain is not locked into a cycle), $\bar{\pi}$ is the long-run fraction of time spent in each state.

We estimate $\bar{\pi}$ by applying the transition matrix to an initial distribution π_0 until the result stops changing:

$$\pi_1 = \pi_0\mathbf{P}, \qquad \pi_2 = \pi_1\mathbf{P} = \pi_0\mathbf{P}^2, \qquad \ldots, \qquad \pi_k = \pi_{k-1}\mathbf{P} = \pi_0\mathbf{P}^k.$$

Power Iteration Algorithm

Initialize: Given an ergodic transition matrix $\mathbf{P}$ with N states, a tolerance $\epsilon > 0$, and a maximum number of iterations T, set $t \leftarrow 1$, $\pi \leftarrow [1/N, \ldots, 1/N]$, and `converged` $\leftarrow$ `false`.

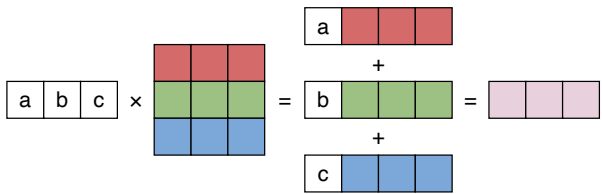
While not `converged` **do**:

1. Compute $\pi' \leftarrow \pi\mathbf{P}$.
2. Check whether to stop:
   - If $\|\pi' - \pi\|_1 \leq \epsilon$ **then** set `converged` $\leftarrow$ `true` and return $\bar{\pi} \leftarrow \pi'$.
   - Otherwise, if $t \geq T$ **then** return π' with `converged` $=$ `false` and a warning.
3. Update the distribution, $\pi \leftarrow \pi'$.
4. Increment the iteration counter, $t \leftarrow t + 1$.

Output: the stationary distribution $\bar{\pi}$, or the last estimate with a warning if the iteration limit was reached.

Inside One Power Iteration Step

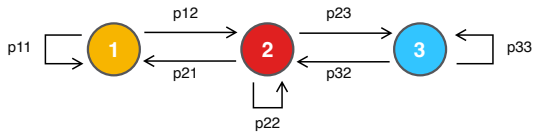
The update $\pi' \leftarrow \pi \mathbf{P}$ is a left vector-matrix product: the new distribution is a weighted sum of the rows of $\mathbf{P}$, weighted by the current probabilities.



- Ergodic: irreducible (every state can be reached from every other state) and aperiodic (not locked into a cycle). An ergodic chain has a unique stationary distribution, and power iteration converges to it from any start.
- Spectral gap: the difference between the largest and second-largest eigenvalue magnitudes of $\mathbf{P}$. Larger gaps converge faster.

Example: A Three-State Markov Chain

In this example, we explore a three-state chain whose states 1 and 3 can reach each other only through state 2:



1. Build the transition matrix and check that every row sums to one.
2. Compute the stationary distribution by power iteration, and by raising $\mathbf{P}$ to a large power, whose rows all approach $\bar{\pi}$.
3. Simulate the chain and compare it with independent draws from $\bar{\pi}$: the state frequencies are similar, but the transitions differ.

Example: Properties of Markov Models and Stationary Distributions

Lab: A Markov Word Generator

In the lab, we build a Markov chain whose states are the letters `a` to `z`:

- we estimate the transition matrix from about 370,000 English words, counting how often each letter follows the first letter of a word,
- we generate new words one character at a time, drawing each character from the row of the character before it,
- we check how many generated words are real, and how that count changes with the length of the word.

Lab: A Markov word generator

Summary

We used Markov models to describe random systems whose future depends only on the present state.

Key takeaways

- **The Markov property.** We describe a random process with one-step transition probabilities instead of its entire history.
- **The transition matrix.** Rows are current states and columns are next states; every entry is non-negative and every row sums to one.
- **The stationary distribution.** For an ergodic chain, power iteration converges to the long-run fraction of time in each state.

Markov models describe random systems in applications ranging from natural language processing to finance and biology.